

GRADUATION PROJECT



Smart IoT-Based Healthcare System

for Early Disease Detection Using Plasmonic PCF-SPR Sensors

Photonic Crystal Fiber – Surface Plasmon Resonance biosensor design optimized with **Machine Learning**, paired with a working **IoT** healthcare prototype



PROJECT TEAM

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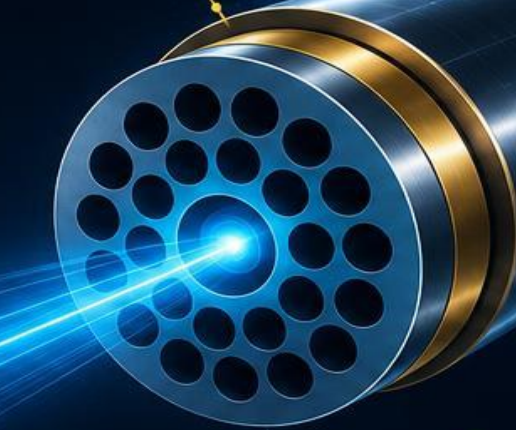


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Gold (Au)
SPR Layer

Surface Plasmon
Resonance



What We'll Cover Today

01



Introduction

Optical sensing, biosensors & SPR fundamentals

02



Literature Review

Prior PCF-SPR designs and the research gap

03



Sensor Design & Results

FEM simulation, optimization & ML validation

04



IoT Healthcare System

Practical ESP32-based monitoring prototype

05



Website Platform

Real-time dashboard & project portal

06



Conclusion & Future Work

Challenges, achievements, what's next

Why Smart Health Monitoring, Why Now

Modern medicine needs fast, accurate tools for early disease detection and continuous patient monitoring. Biosensors — especially those based on Surface Plasmon Resonance (SPR) — detect biological change with remarkable sensitivity. Paired with the Internet of Things (IoT), they enable real-time, remote healthcare.



High Sensitivity

SPR detects tiny refractive-index shifts from biological changes



Real-Time & Remote

IoT enables continuous monitoring beyond the clinic



This Project

Combines an optical biosensor design with a working IoT prototype

Limitations of Conventional Sensing



Low Sensitivity

Cannot detect small changes in biological samples



Limited Range

Narrow detection window misses many biomarkers



Large & Bulky

Difficult to use in clinical or portable settings



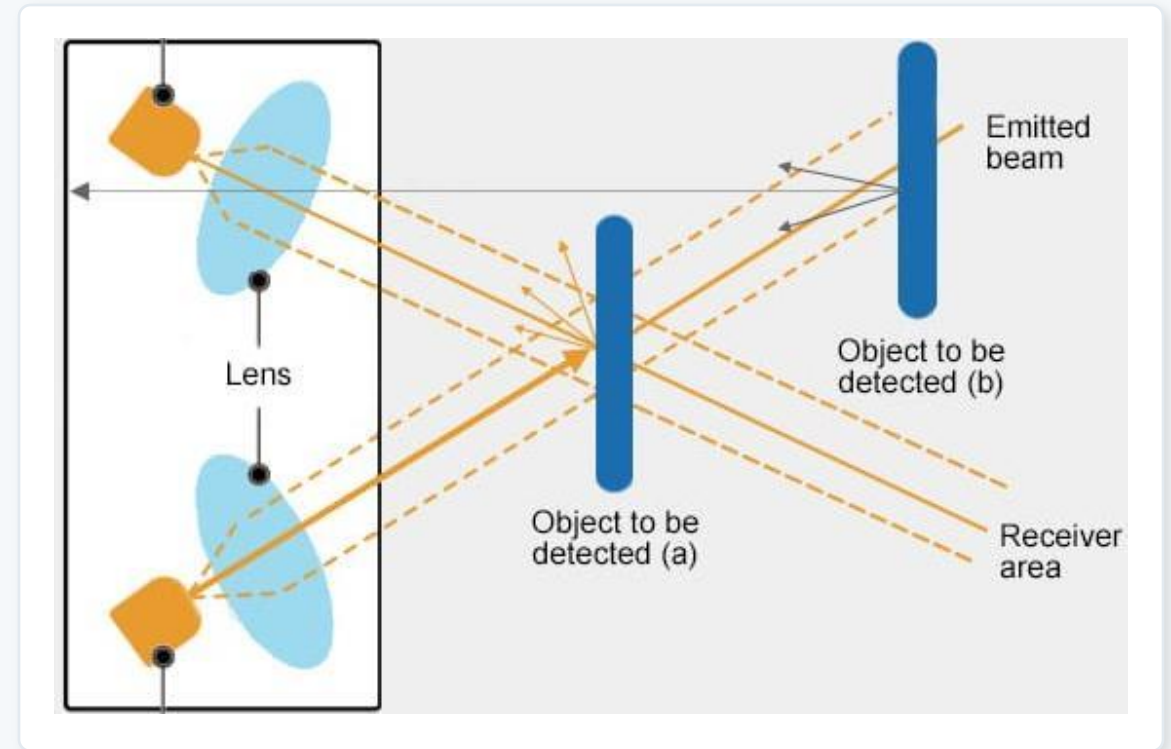
Noise-Sensitive

Susceptible to environmental & electromagnetic noise

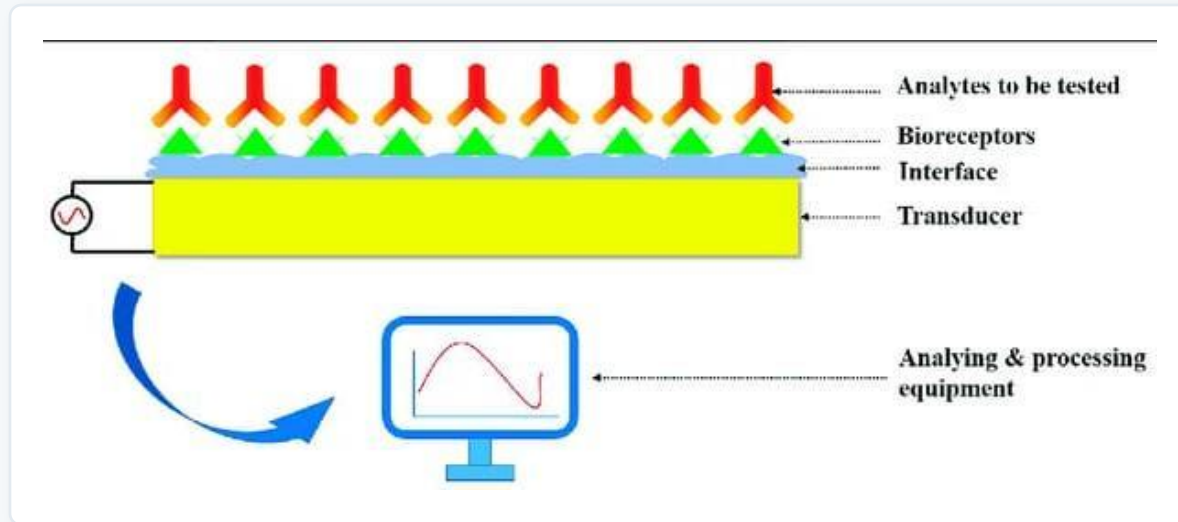
These limitations make accurate, real-time medical detection difficult — motivating the shift toward optical sensing technologies such as Surface Plasmon Resonance.

Optical Sensing: The Core Principle

- Detects physical, chemical or biological change by measuring variations in light: intensity, wavelength, phase or polarization
- Immune to electromagnetic interference, unlike electronic sensors
- Enables label-free detection with fast response times
- Foundation for advanced techniques used later in this project: SPR, interferometry, PCF sensing



Biosensors: Structure & Types



A biosensor pairs a biological recognition element (enzymes, antibodies, DNA, cells) with a transducer that converts the biological reaction into a measurable electrical or optical signal.

TYPES OF BIOSENSORS

Enzymatic biosensors

Immunosensors (antigen–antibody)

DNA biosensors

Cell-based biosensors

Optical biosensors (e.g. SPR)

Surface Plasmon Resonance (SPR)

SPR occurs when incident light interacts with free electrons at a metal–dielectric interface, generating surface plasmons that are extremely sensitive to refractive-index changes nearby.

Prism-based

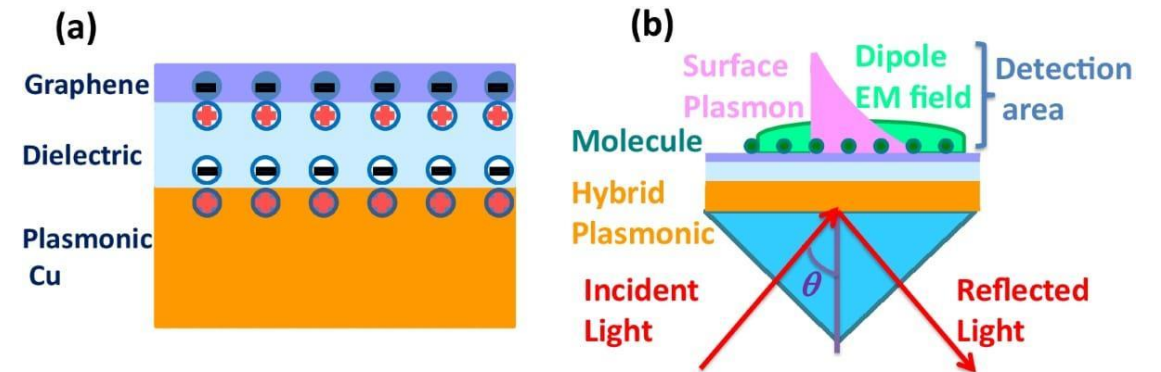
High precision, lab-scale

Grating-based

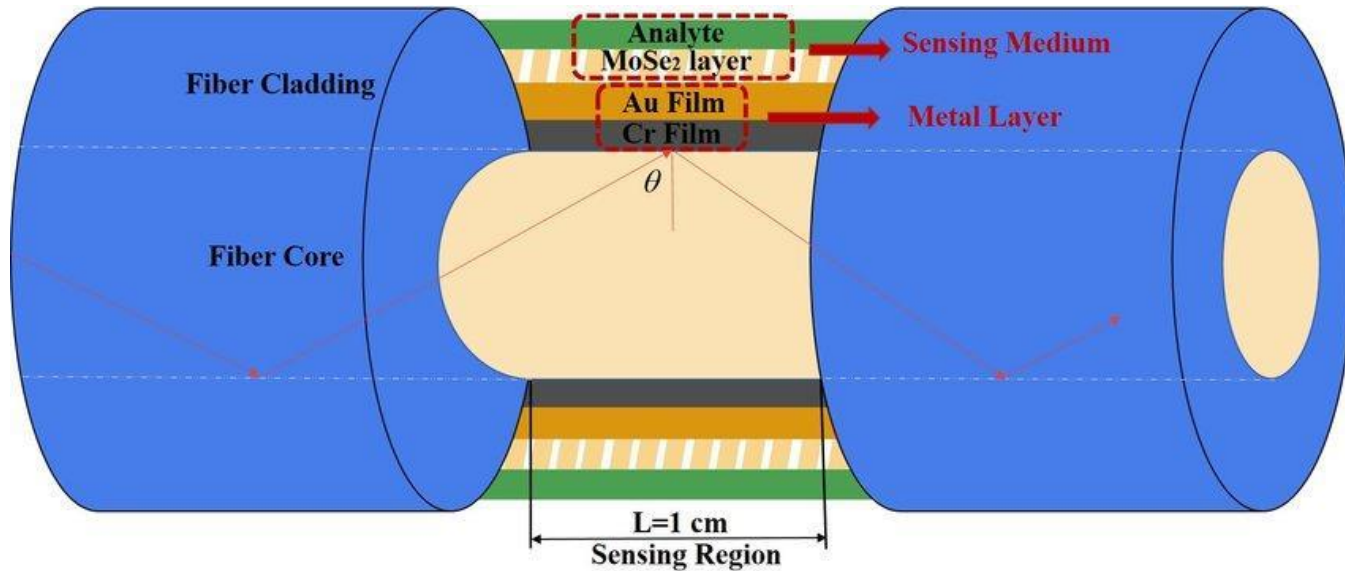
Compact, portable platforms

Fiber-optic

Remote sensing, harsh environments



Fiber-Optic SPR Sensors



Light travels through the fiber and interacts with a metal-coated region, exciting surface plasmons. Resonance is highly sensitive to the refractive index of the surrounding medium.



Compact & flexible



Remote sensing capability



Compatible with harsh environments

Photonic Crystal Fiber (PCF)

A Photonic Crystal Fiber is an optical fiber with a periodic arrangement of air holes running along its length — giving precise control over light confinement, propagation and dispersion.



Index-Guiding & Bandgap Types

Light confined by a solid high-index core, or by photonic bandgap effects



Tunable Hole Geometry

Hexagonal, square & hybrid arrangements tailor dispersion & birefringence



High Light Confinement

Low transmission loss with controllable dispersion



Ideal for Sensing

Enhanced light–analyte interaction; compatible with plasmonic coatings

PCF-SPR: Why Combine Them?

Engineered air-hole geometries enhance evanescent-field interaction and phase matching — giving PCF-SPR sensors superior performance over conventional fiber sensors, with real-time, label-free detection.



Biosensing

Glucose, proteins, DNA, cancer biomarkers



Chemical Sensing

Toxic chemicals & industrial pollutants



Environmental Monitoring

Water pollutants & heavy-metal ions



Medical Diagnostics

Early disease detection, point-of-care testing

Project Objectives

1

Design a high-performance PCF-SPR biosensor for detecting small refractive-index variations

2

Analyze optical characteristics using the Finite Element Method (FEM) in COMSOL Multiphysics

3

Investigate the effect of air-hole diameter, pitch & gold-layer thickness on performance

4

Optimize the design using AI: Machine Learning and Genetic Algorithm techniques

5

Evaluate wavelength sensitivity, amplitude sensitivity & sensing resolution

6

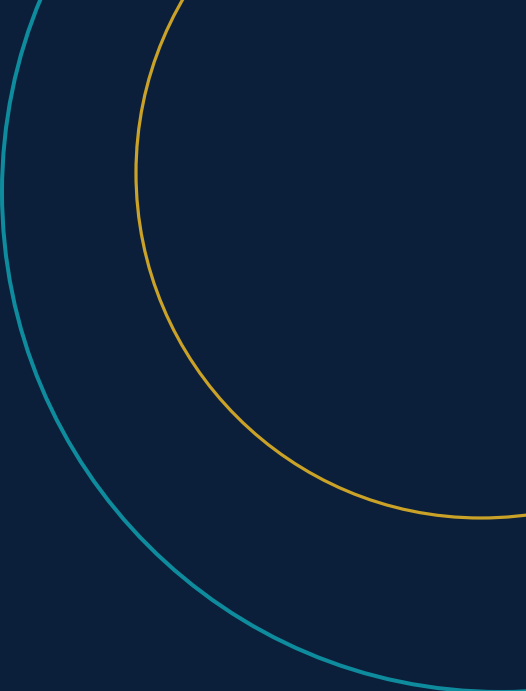
Develop a practical IoT healthcare monitoring prototype using ESP32 + biomedical sensors

7

Implement real-time data transmission & visualization via Blynk and Firebase

8

Combine optical sensor theory with a working smart-healthcare IoT system

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CHAPTER 2

Literature Review

Learning from a decade of PCF-SPR sensor research to find the gap this project addresses.

How Recent PCF-SPR Designs Compare

Ref.	Core Type	RI Range (RIU)	Sensitivity (nm/RIU)	Stability	Fabrication
[10] 2025	Single-core	1.25–1.43	47,000	Medium	High
[13] 2025	Single-core	1.36–1.42	21,250	Medium	High
[14] 2023	Single-core	1.32–1.42	77,500	Medium	High
[11] 2025	Single-core	1.33–1.39	6,000	Medium	Medium
[16] 2022	Single-core	1.33–1.39	6,000	Medium	Medium
[18] 2024	Single-core	1.33–1.39	4,800	Low	Medium
[12] 2018	Single-core	1.36–1.38	3,340	Low	Medium
Proposed [20]	Multi-core/channel	1.21–1.42	5,000 (up to 8,000)	High	Low–Medium

Most prior designs push sensitivity via highly complex geometries. The proposed sensor instead balances sensitivity, stability and fabrication feasibility — the practical gap this project fills.

What This Project Adds



Balanced performance

Competitive sensitivity without relying on extreme, hard-to-fabricate geometries



Simpler, robust structure

Circular geometry keeps fabrication feasibility high



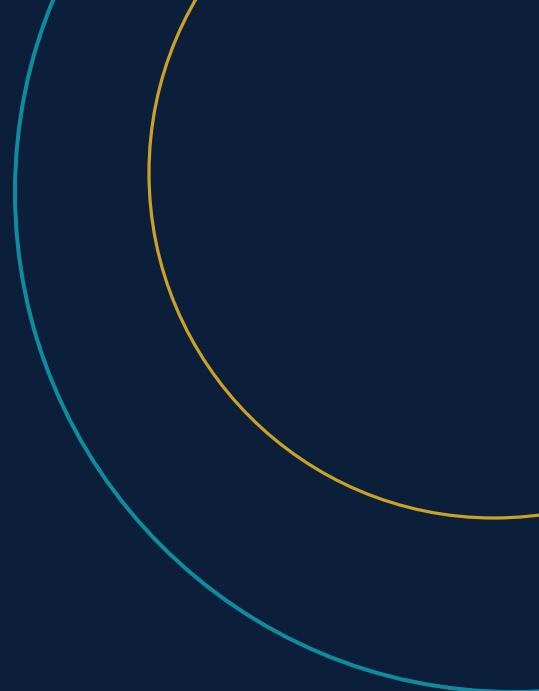
Multi-channel capability

Improved stability and reproducibility for real biosensing use



AI-assisted optimization

Machine Learning + Genetic Algorithm push performance further, efficiently

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CHAPTER 3

Sensor Design, Simulation & Performance

Finite Element simulation in COMSOL Multiphysics, followed by AI-driven optimization.

Proposed Sensor Geometry

A circular photonic crystal fiber design: 15 air holes arranged symmetrically around a central core, with a thin gold (Au) layer generating the SPR effect.

Silica (SiO₂) body

Low loss, high mechanical stability

15 air holes

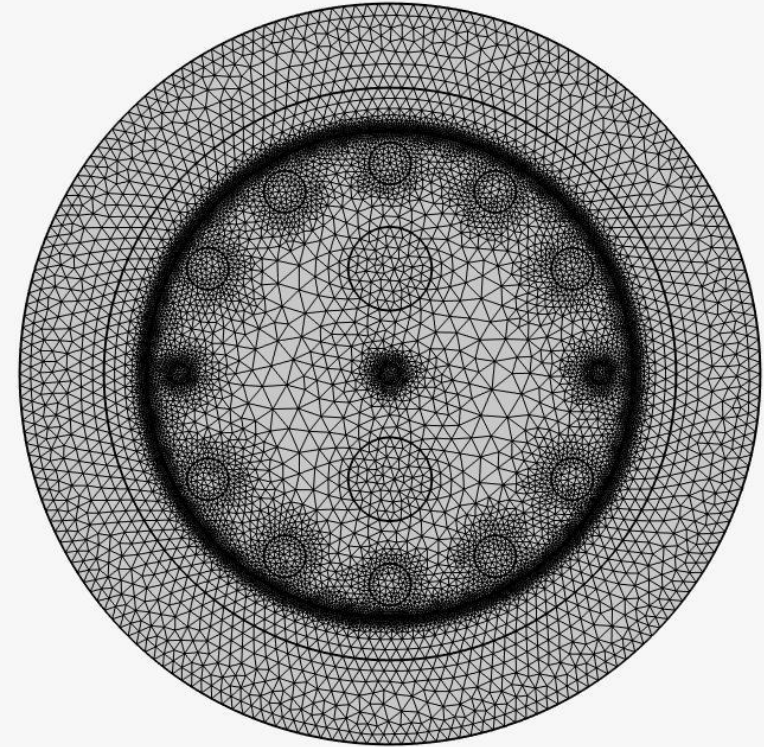
$d_1 = 1.6 \mu\text{m}$, $d_2 = 0.4 \mu\text{m}$, $d_3 = 0.8 \mu\text{m}$

Gold (Au) sensing layer

Chemically stable, strong plasmonic response

Circular symmetric layout

Balances confinement & evanescent-field reach



Numerical Analysis: Finite Element Method

Maxwell's equations are solved across the fiber cross-section using FEM in COMSOL Multiphysics, converting the wave equation into an eigenvalue problem to find propagation constants and field distributions.



High accuracy for complex, curved geometries



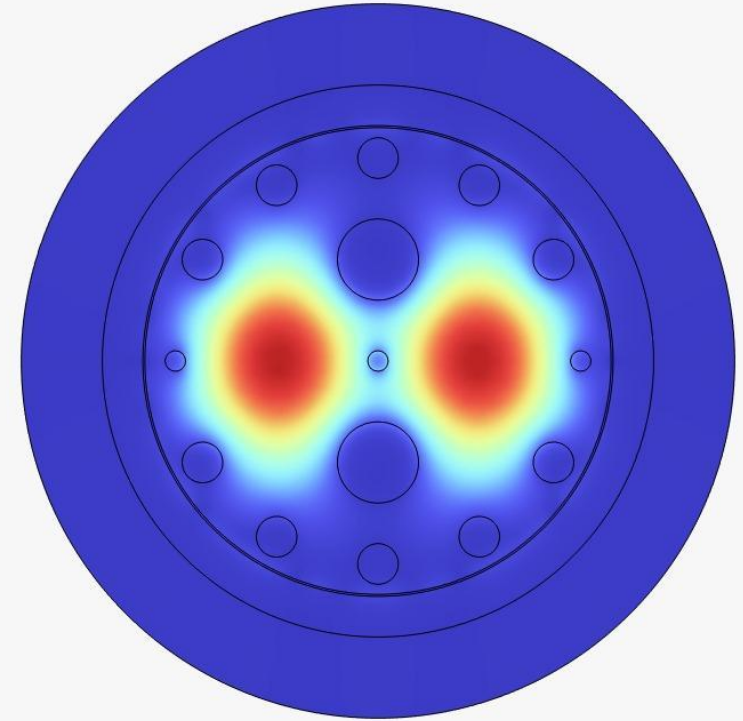
Precisely models thin gold layers



Accurate E/H field & confinement-loss results

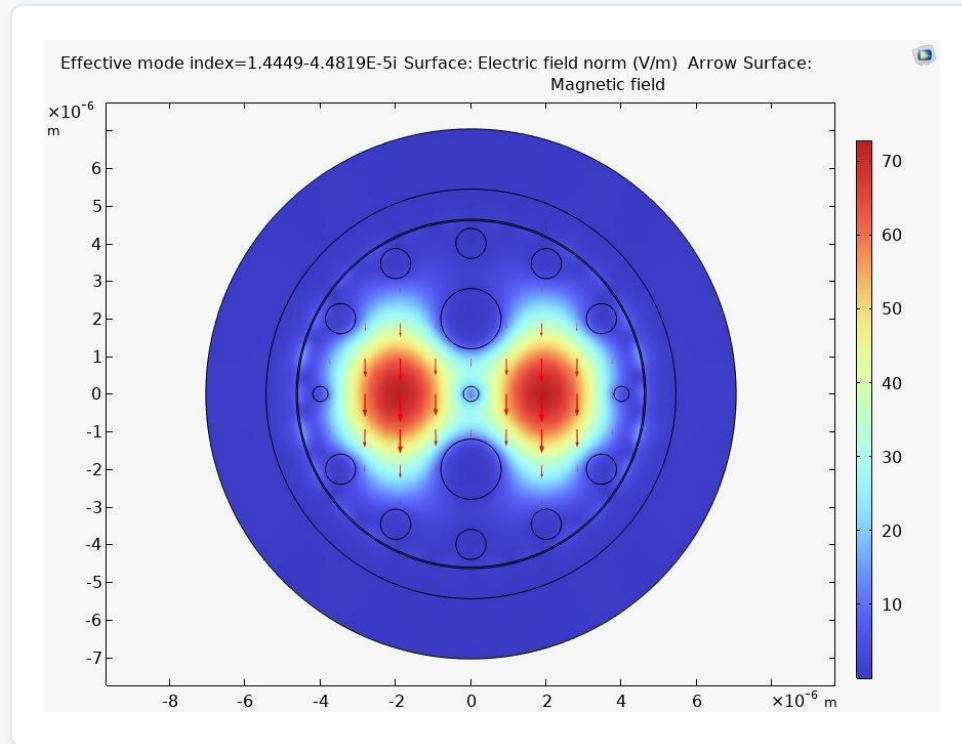


Perfectly Matched Layer (PML) minimizes boundary reflection

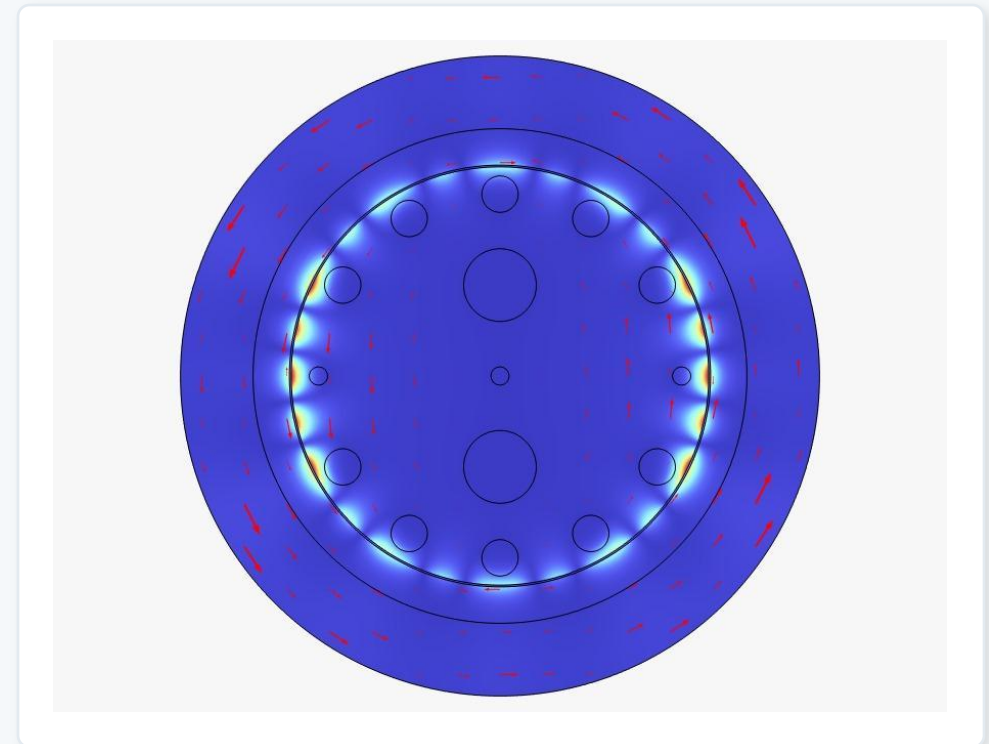


Mesh: ~61,250 elements • Numerical error $\approx 0.12\%$ • Finer mesh near the gold layer for SPR accuracy

Fundamental Core Mode & Field Distribution



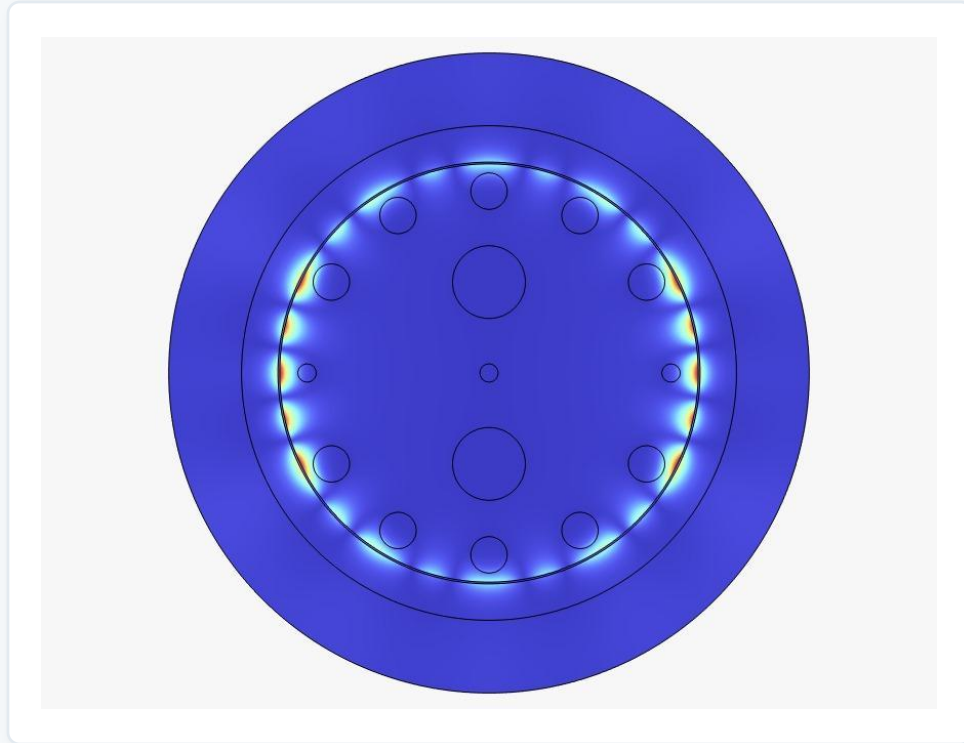
Fundamental Core Mode



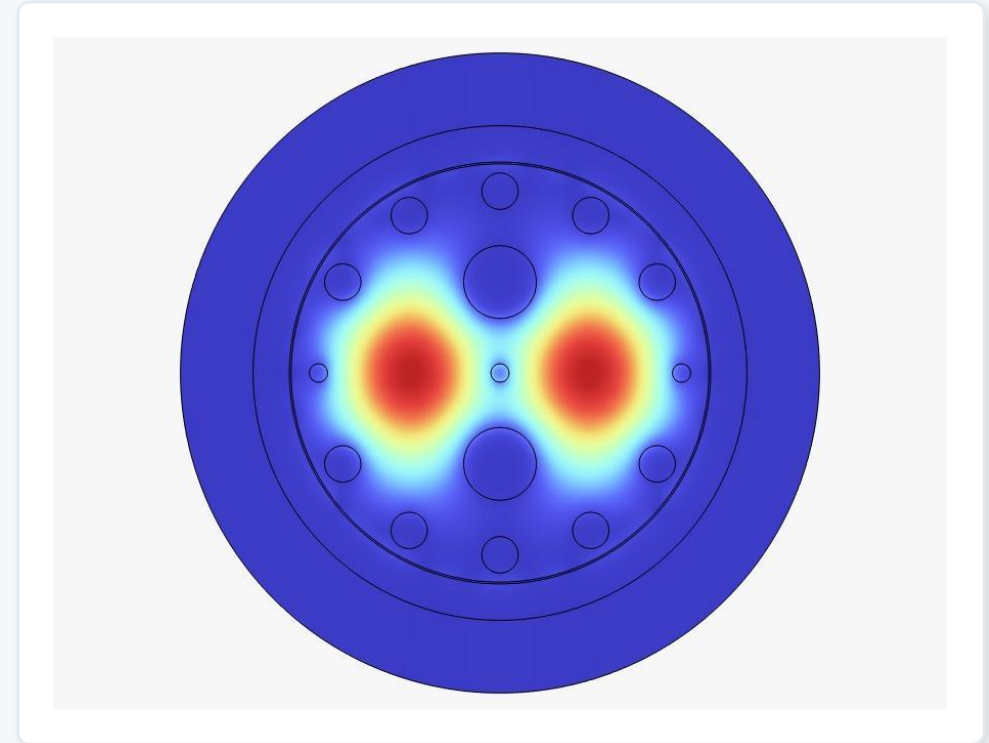
Electric & Magnetic Field Distribution

Light stays strongly confined within the core (two symmetric field lobes) — negligible power reaches the gold layer, confirming no coupling before resonance.

Surface Plasmon Polariton (SPP) Mode



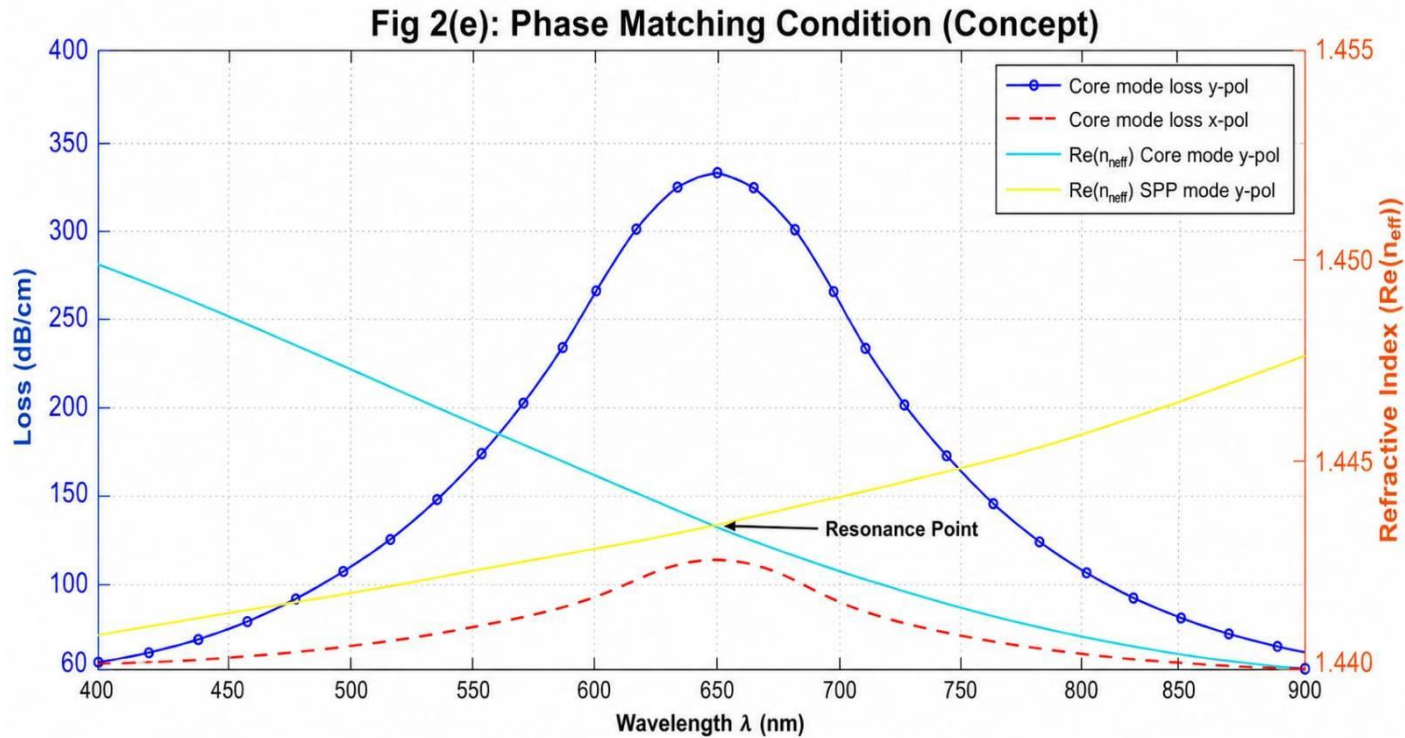
SPP Mode: Field at the Gold Layer



Core Mode for Comparison

At resonance, the field concentrates strongly around the gold interface — proof that the guided mode has coupled into a surface plasmon polariton, confirming SPR excitation.

Phase Matching Condition



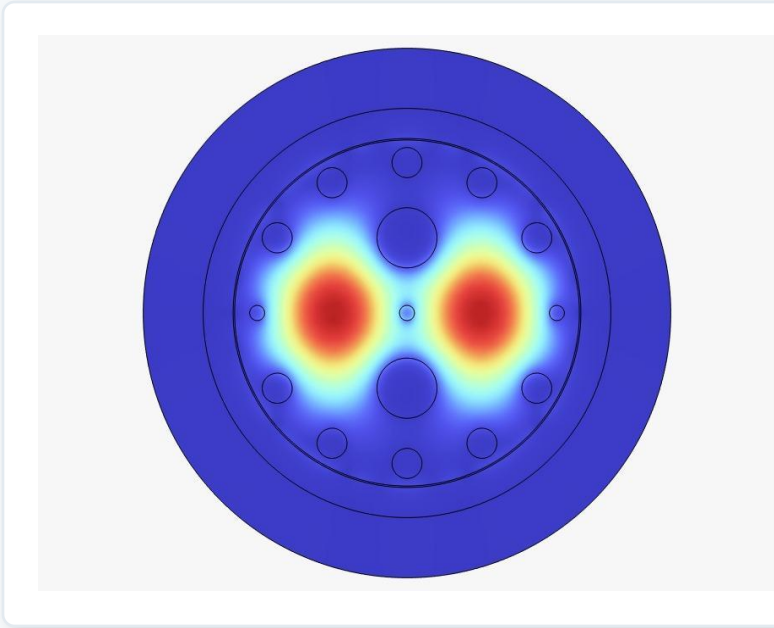
Where Core Mode Meets SPP Mode

As wavelength increases, the core mode's effective index falls while the SPP mode's rises. They intersect near 650 nm — the phase-matching (resonance) point — producing a sharp confinement-loss peak.

$\approx 650 \text{ nm}$

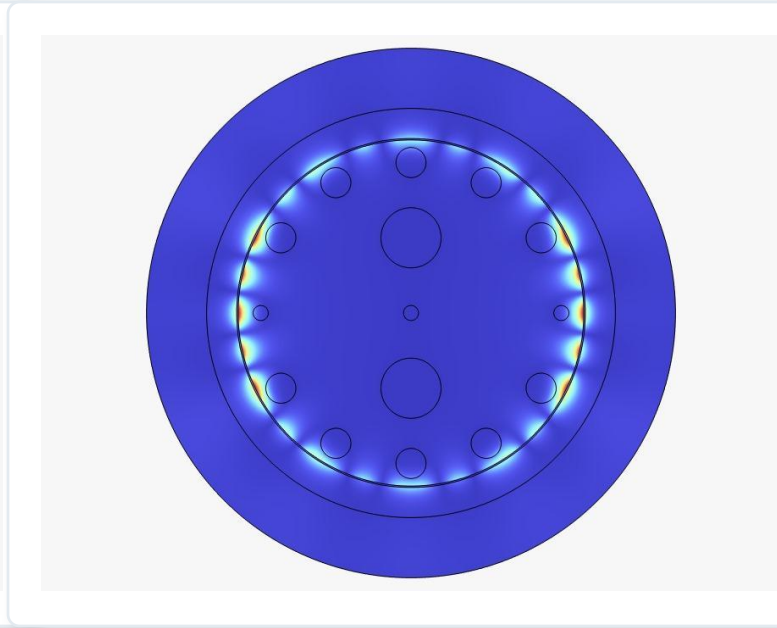
Resonance wavelength where phase matching occurs

Energy Transfer: Three Stages



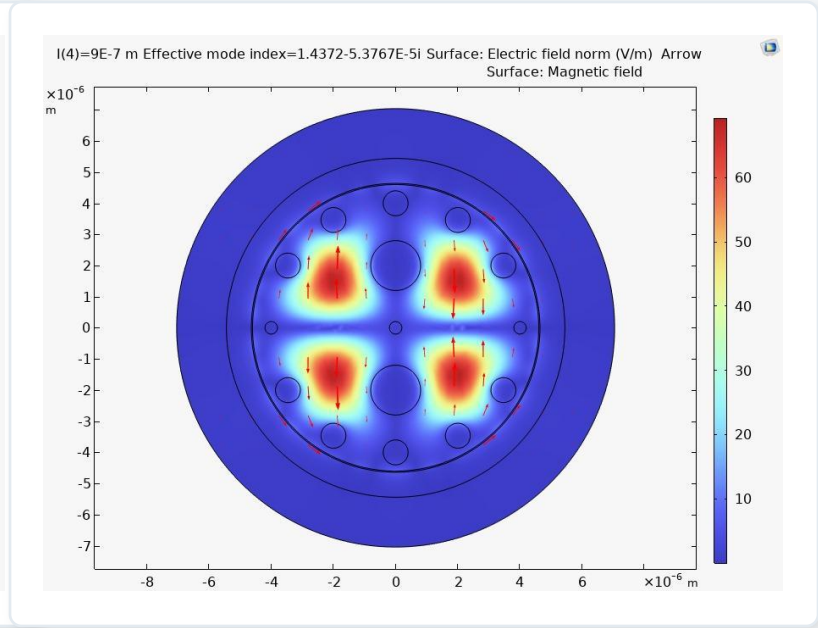
Before Resonance

Field confined in core; negligible loss



At Resonance

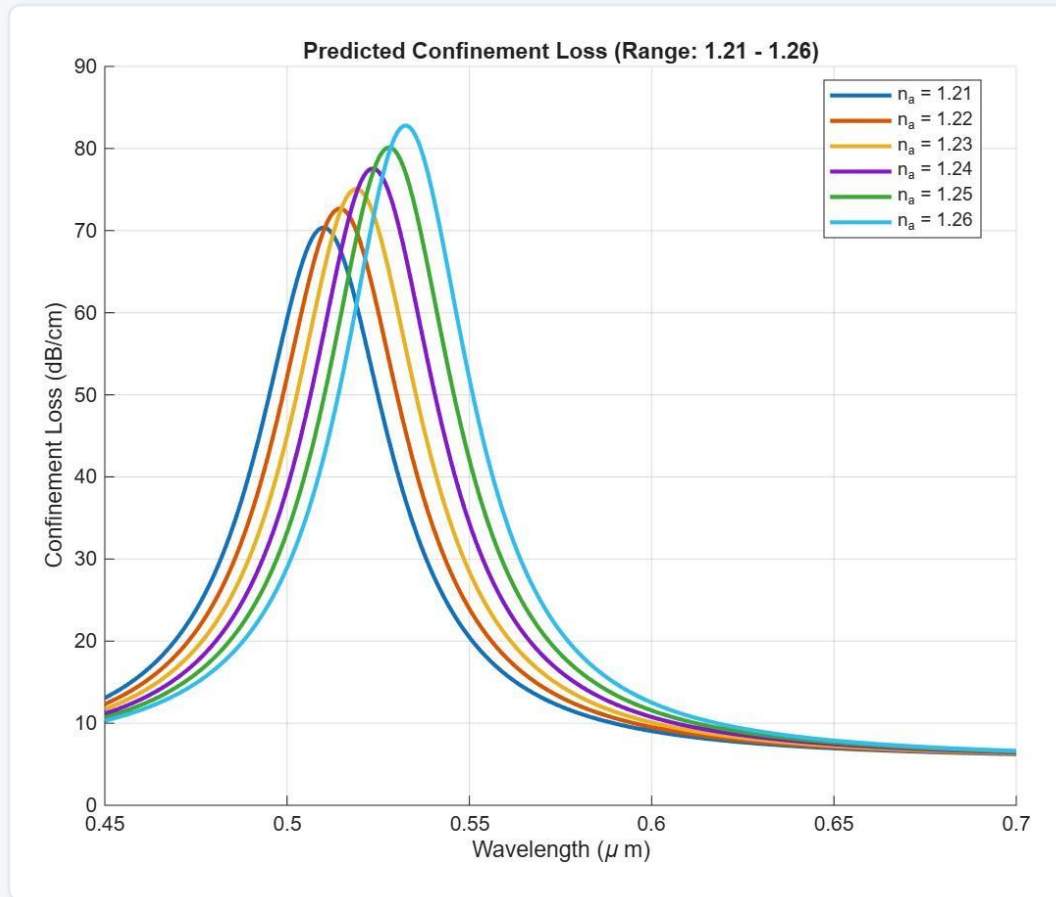
Maximum coupling; energy peaks at gold layer



After Resonance

Modes diverge again; field returns to core

Confinement Loss Spectrum vs. Refractive Index



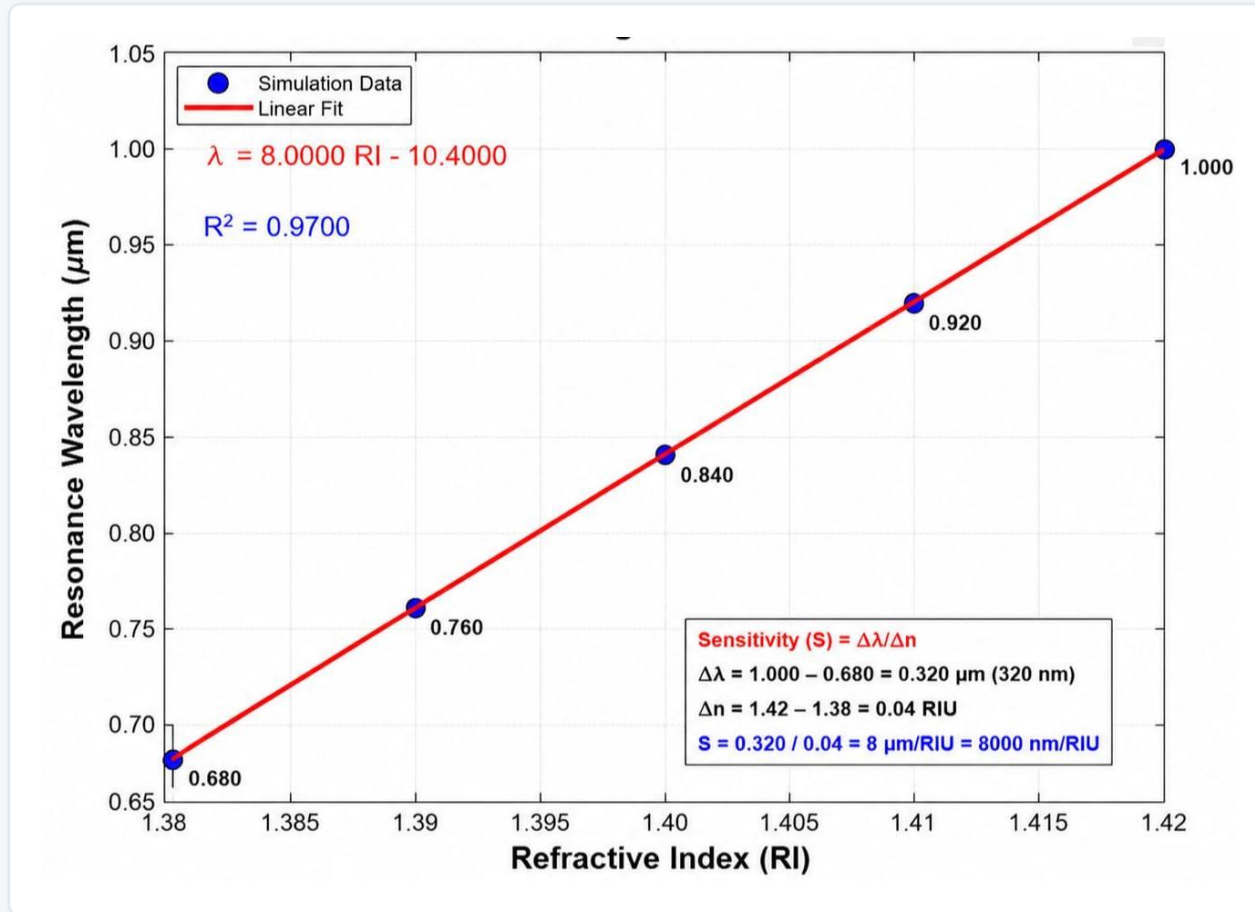
The Red-Shift Signature

As analyte refractive index increases, each resonance peak shifts to longer wavelengths. Tracking this shift — wavelength interrogation — is the sensor's fundamental sensing mechanism.

1.38 → 1.42

Refractive index sensing range (RIU)

Resonance Wavelength vs. Refractive Index



$$R^2 = 0.97$$

Excellent linear fit — simplifies calibration

320 nm

Total resonance shift across the tested RI range

8000 nm/RIU

Peak wavelength sensitivity achieved

Wavelength Sensitivity Analysis

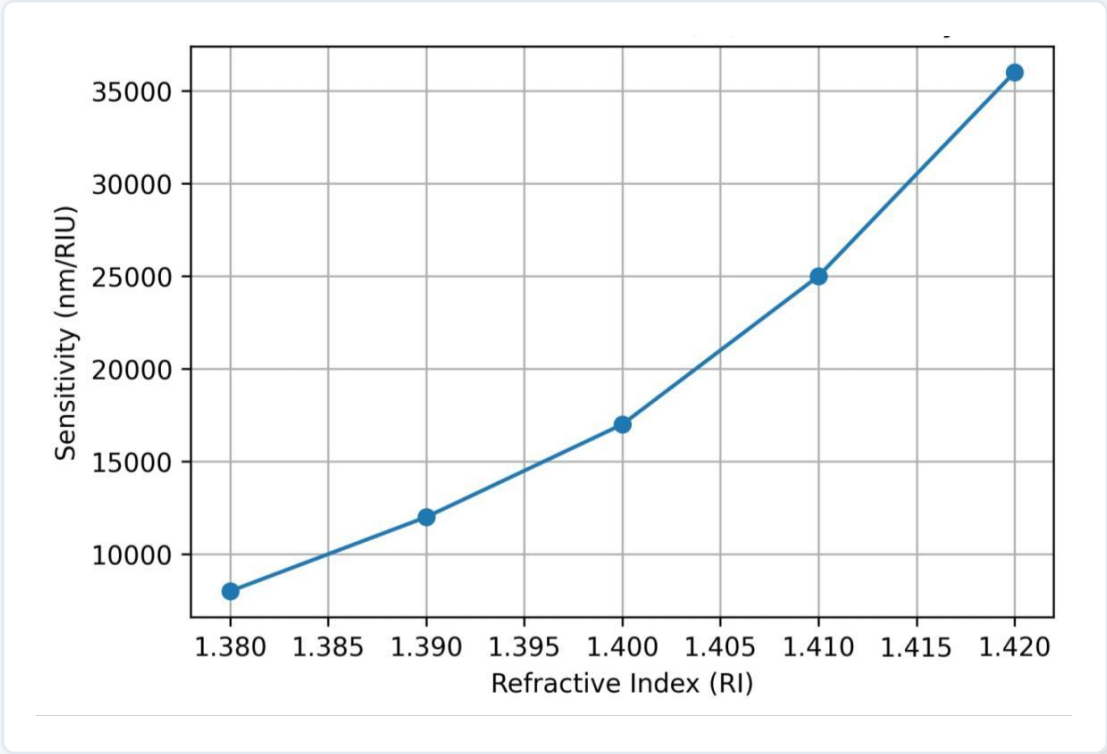
Sensitivity is the resonance-wavelength shift per unit change in analyte refractive index. As RI rises, stronger SPP coupling produces a larger shift for the same RI change — sensitivity increases with RI.

8000

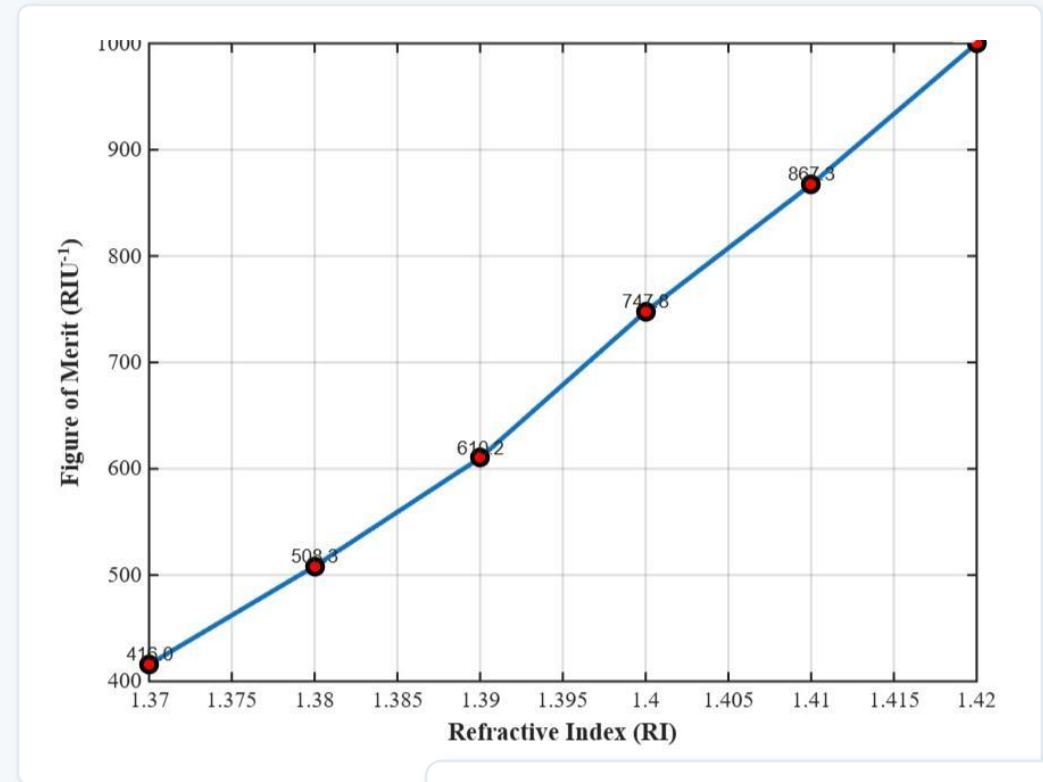
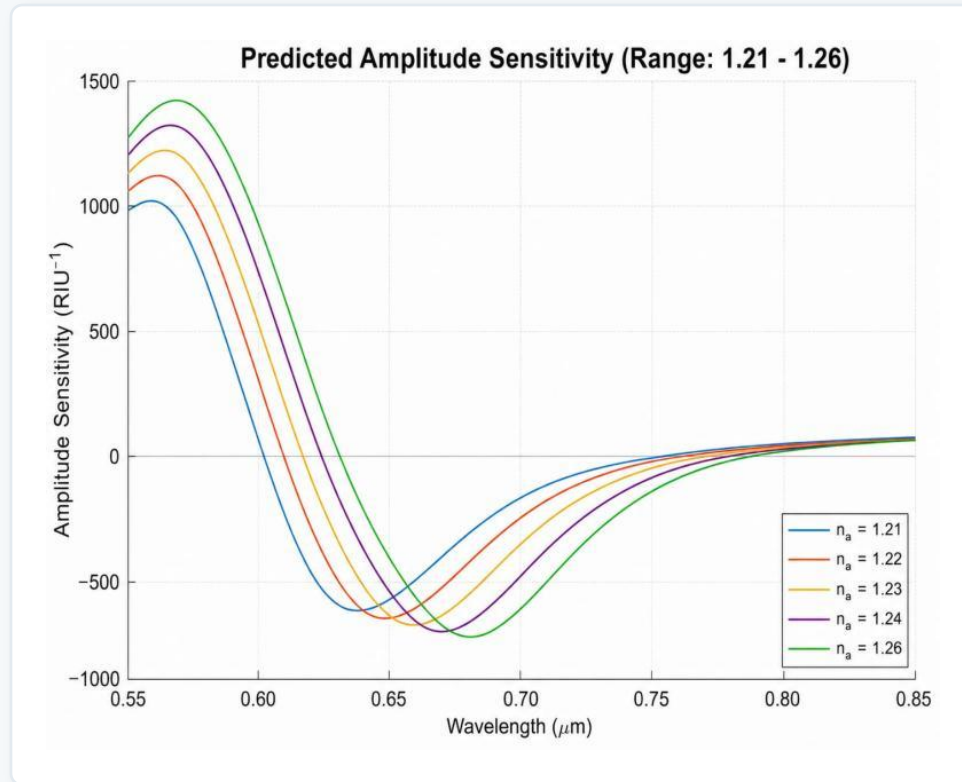
nm/RIU — maximum wavelength
sensitivity

1.38–1.42

RIU — the tested sensing range



Amplitude Sensitivity & Figure of Merit



Amplitude sensitivity tracks the relative change in confinement loss; FOM = sensitivity $\div$ resonance linewidth (FWHM) — higher FOM means sharper, more resolvable resonance peaks.

2600 RIU^{-1}

Max. amplitude sensitivity $\cdot$ FOM up to
~1000 RIU^{-1}

Detection Limit (DL)

Sensitivity	FWHM	FOM	Performance
High	Small	High	Excellent
High	Large	Medium	Good
Low	Small	Medium	Acceptable
Low	Large	Low	Poor

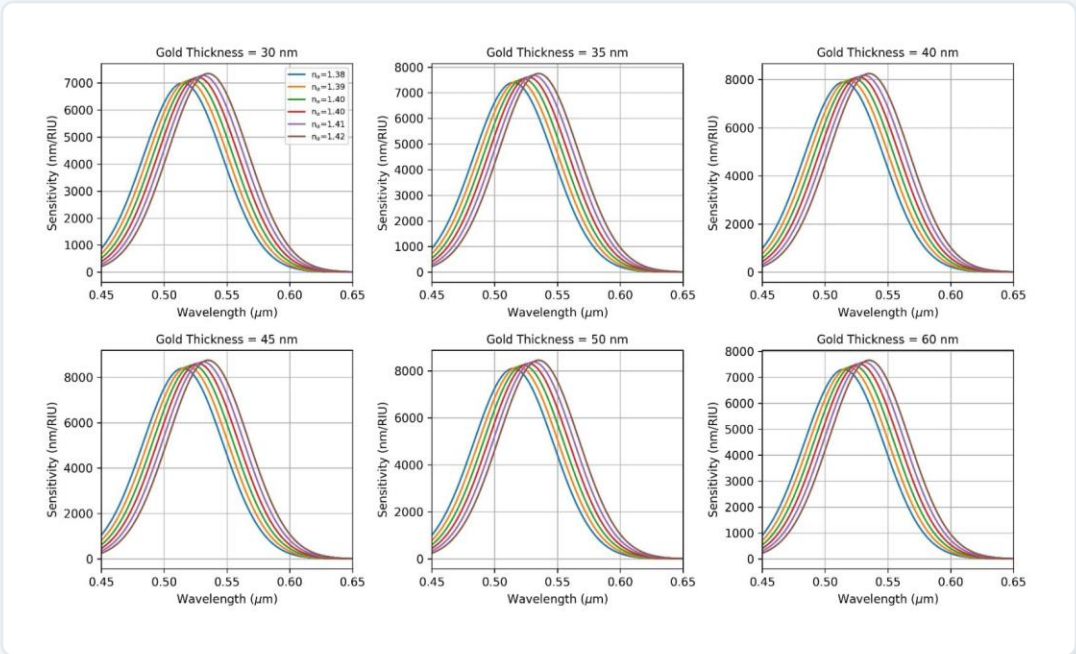
Detecting the Smallest Changes

The detection limit falls as RI rises from 1.37 to 1.42 — because DL is inversely proportional to sensitivity. Higher sensitivity at higher RI directly improves detection accuracy.

1.9×10^{-5} RIU

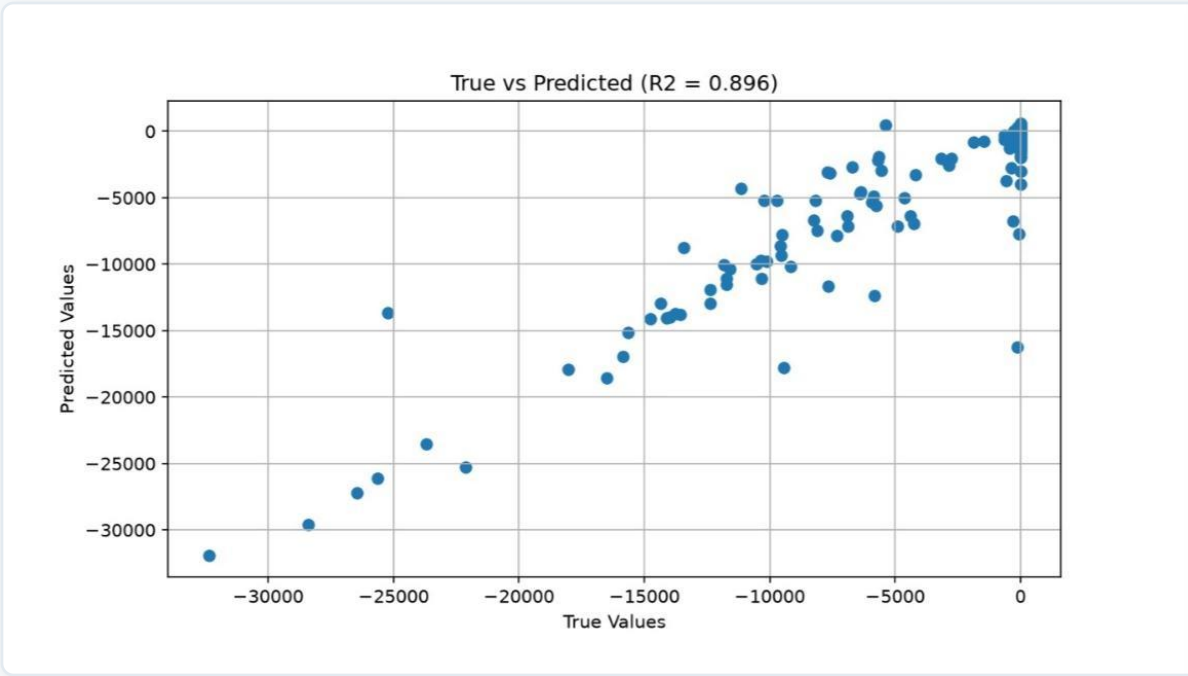
Best-case sensor resolution (at RI = 1.42, $\sim 8.93 \times 10^{-6}$ RIU)

Parametric Optimization Results

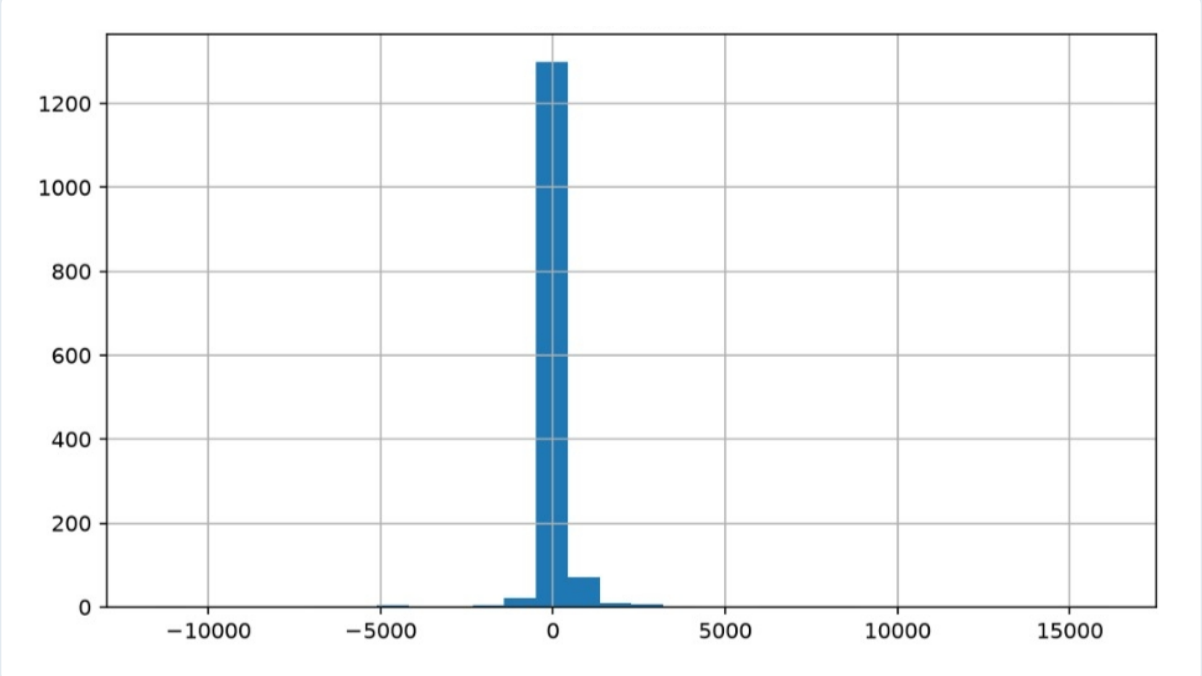


Parameter	Initial Value	Optimized Value	Improvement
Gold Thickness	30 nm	40 nm	87.5%
Air-Hole Diameter	600 nm	850 nm	71.4%
Pitch (Λ)	1.8 μm	2 μm	94.6%

Machine Learning: Prediction & Validation



True vs. Predicted Sensor Response



Prediction Error Distribution

Comparison with Published Work

	Target	Background	Wavelength Sens.	Detection Limit	Amplitude Sens.
Roni et al. (2020) [20]	Blood / cancer skin	Silica	270 nm/RIU	2×10^{-5} RIU	1277.47
Proposed Sensor	Blood / cancer skin	Silica	330 nm/RIU	1.9×10^{-5} RIU	2600

The proposed sensor improves on the closest published reference across every key metric — wavelength sensitivity, detection limit, and amplitude sensitivity — while remaining practically fabricable.

The Optimized Sensor, at a Glance

8000

nm/RIU
Max. wavelength sensitivity

2600

RIU⁻¹
Max. amplitude sensitivity

1.9×10^{-5}

RIU
Sensor resolution

0.97

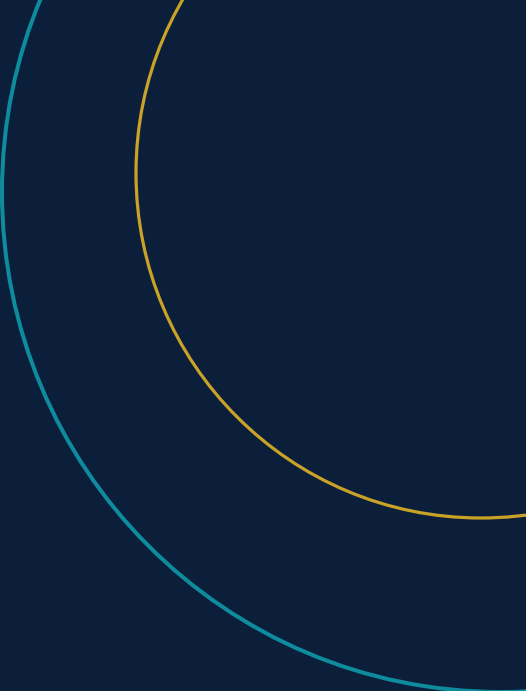
R²
Resonance-wavelength linearity

0.896

R²
Machine-learning prediction accuracy

40 nm / 850 nm / 2 μ m

Optimized gold thickness, hole diameter, pitch

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CHAPTER 4

From Optical Theory to Practical IoT System

Why we built a low-cost electronic prototype alongside the optical sensor design.

Why an IoT Alternative?

The PCF-SPR sensor achieved excellent sensitivity — but practical fabrication faces real barriers:

-  Precise photonic-crystal-fiber fabrication
-  Nanometer-scale gold-layer deposition
-  High-cost optical components
-  Specialized spectroscopy equipment required
-  Difficult to use outside the laboratory

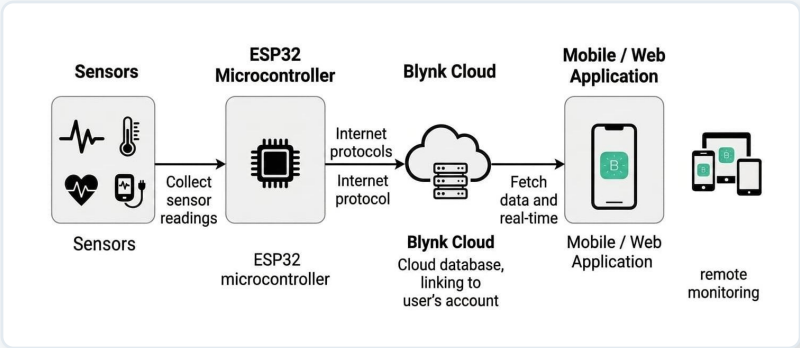
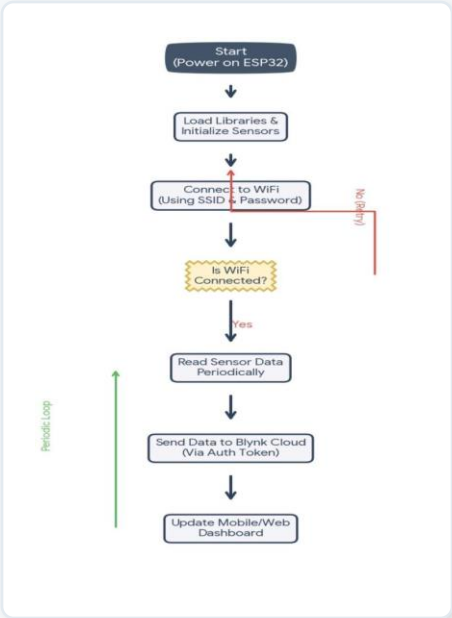
Feature	PCF-SPR	Proposed IoT System
Cost	High	Low
Fabrication	Complex	Simple
Portability	Limited	High
Real-Time Monitoring	Moderate	Excellent
Maintenance	Difficult	Easy

Proposed IoT Healthcare System

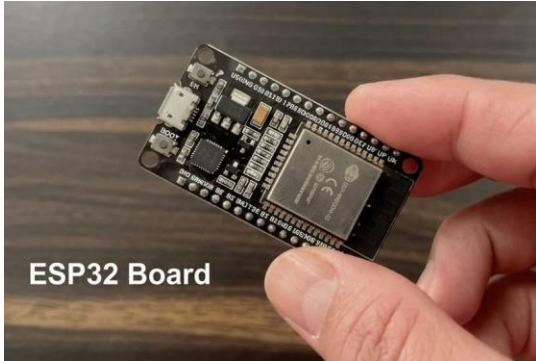


System Workflow

Power on → connect WiFi → read sensors periodically → transmit to Blynk Cloud via auth token → dashboard updates in real time on mobile & web.



The Sensor & Control Components



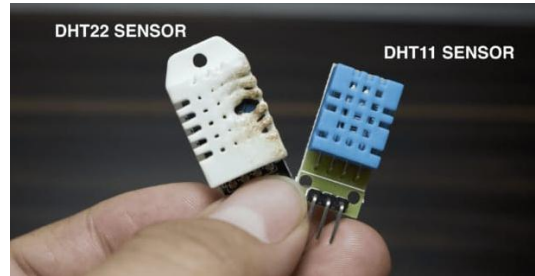
ESP32 Microcontroller

Built-in WiFi + Bluetooth, dual-core, low power



MAX30102 Sensor

Heart rate & SpO₂ via photoplethysmography (I²C)



DHT11 Sensor

Ambient temperature & humidity



DS18B20 Sensor

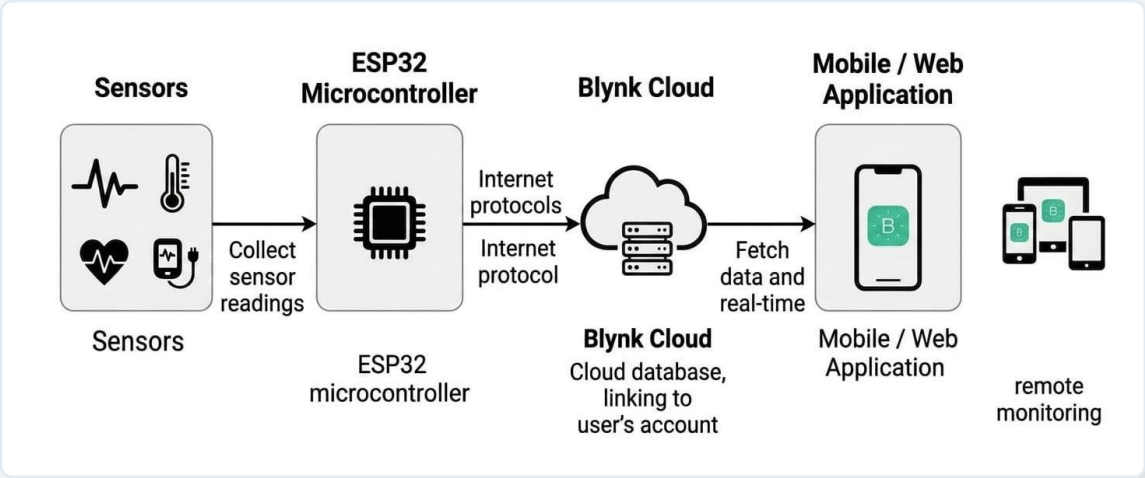
Precise body temperature (One-Wire, -55 to +125°C)

Sensors → ESP32 → WiFi → Blynk Cloud / Firebase → Mobile App & Web Dashboard

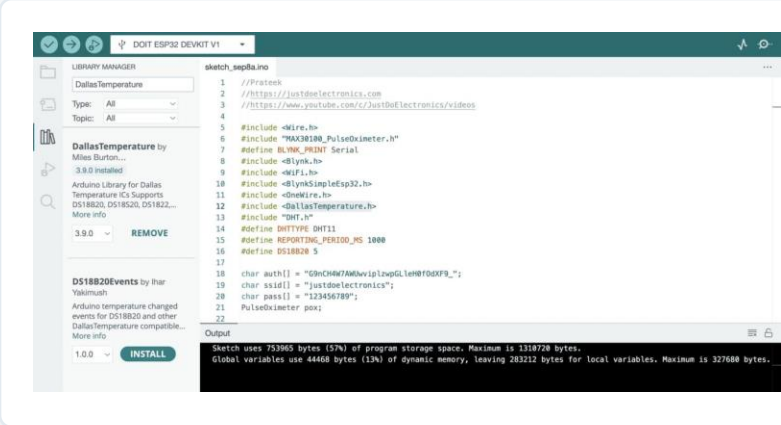
Data in the Cloud

A dedicated Firebase Realtime Database stores every sensor reading, giving the system permanent, remotely accessible storage alongside Blynk's live mobile dashboard.

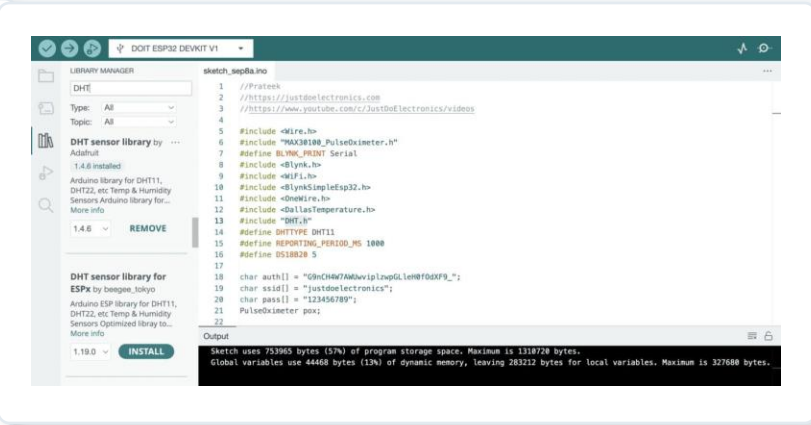
Database URL	doctorr-0123-default-rtdb.firebaseio.com
Storage Bucket	Patient_Data
Access	Web dashboard & mobile app, from anywhere online



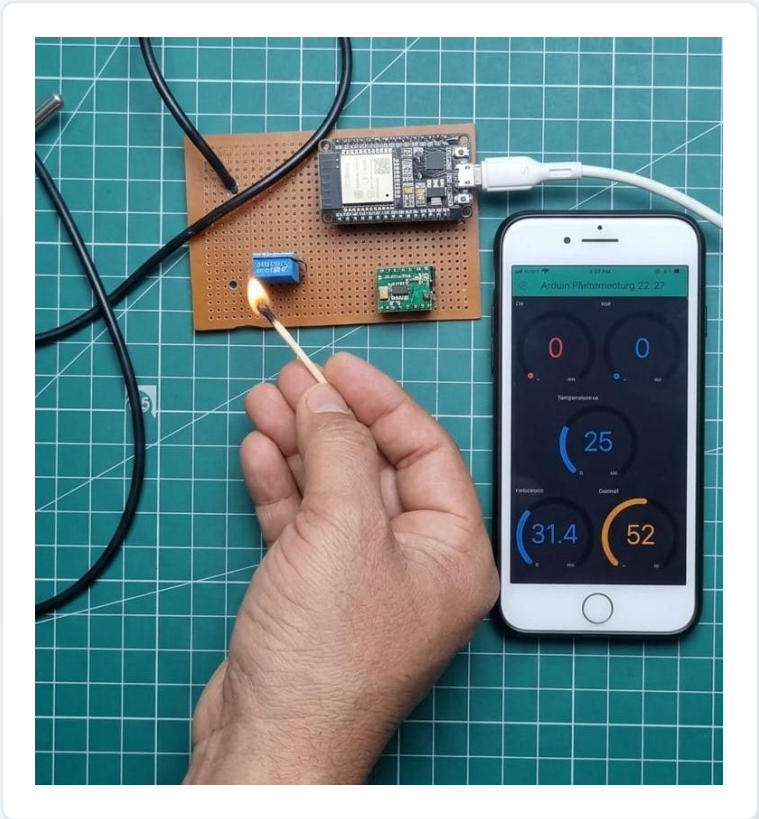
Practical Testing Results



Heart Rate & SpO₂

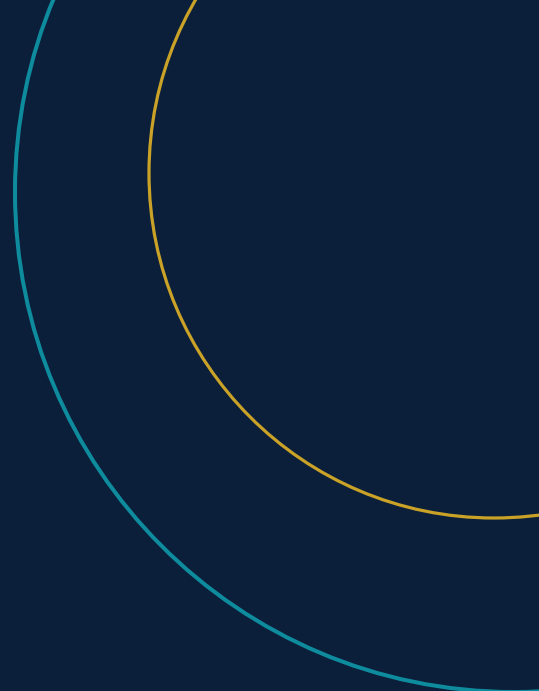


Body Temperature



Live System on the Bench

All three sensors transmitted data reliably to Blynk in real time, confirmed on both the mobile app and web dashboard.

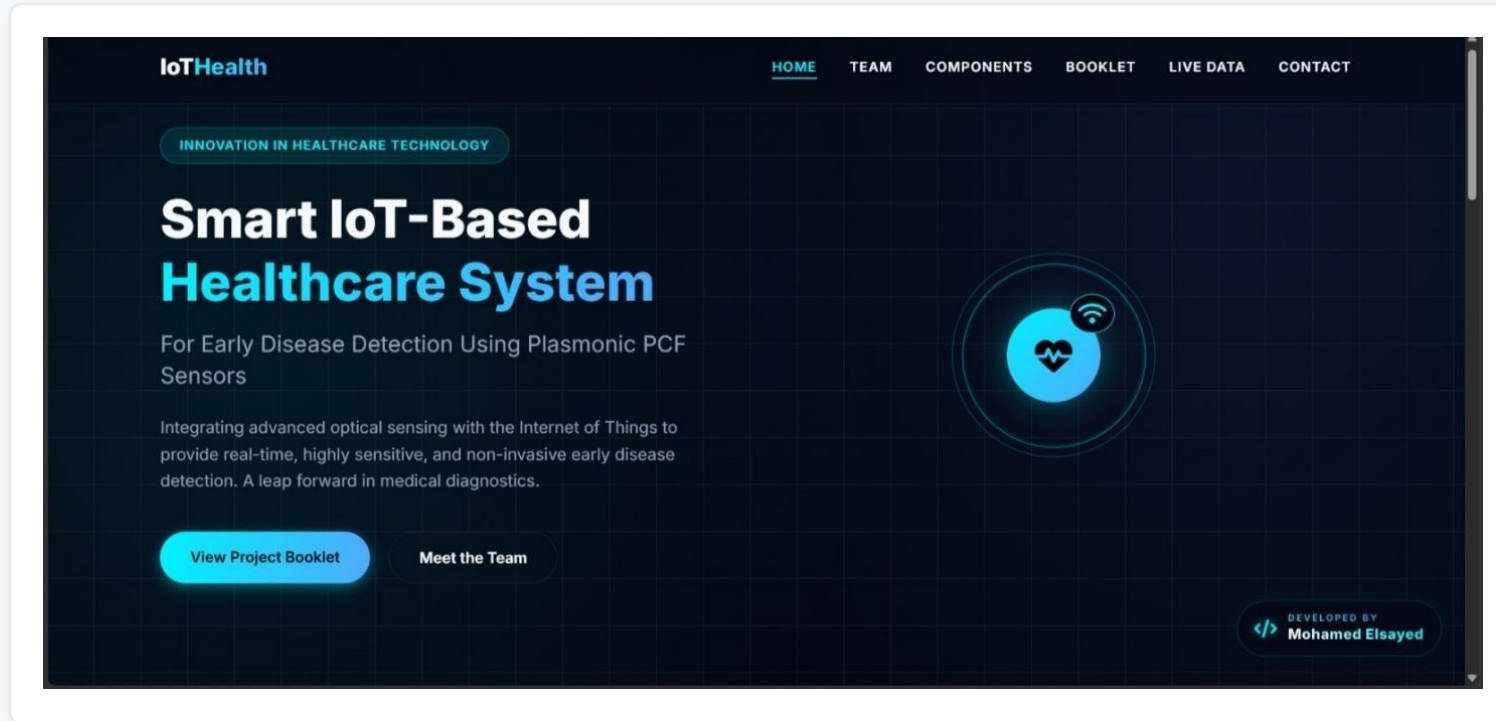


CHAPTER 5

Website Design & Development

A responsive web platform connecting patients, physicians and the project itself.

The Smart IoT Healthcare Website



Built With

- ✓ HTML5, CSS3, JavaScript (ES6)
- ✓ Responsive Web Design
- ✓ Speech Synthesis API
- ✓ Intersection Observer API

From Live Data to AI-Assisted Reports



Live Data Dashboard

Real-time vitals for fast, remote monitoring



AI Diagnostic Report

Flags abnormal readings for quick decisions



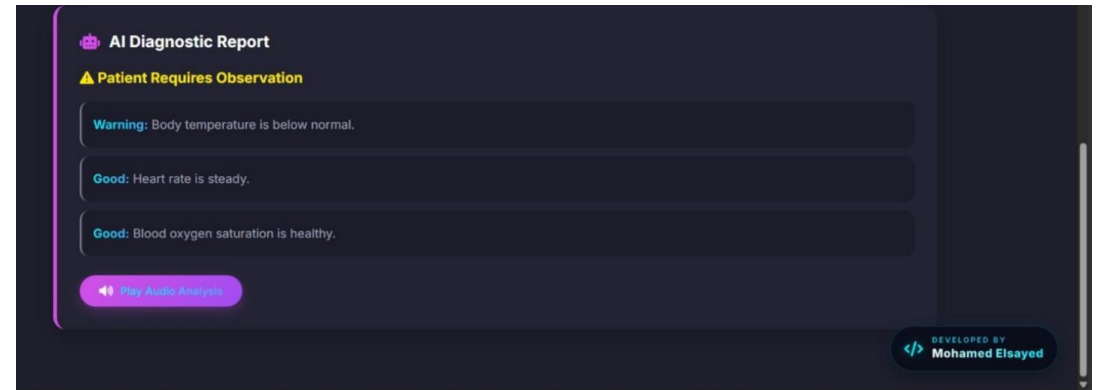
Team Page

Interactive cards; pronounces names via Speech Synthesis



Booklet & Contact

Digital thesis access + direct inquiry form



Challenges We Faced



Optical Fabrication

Precision fabrication of PCF-SPR requires equipment beyond standard labs



Parameter Selection

Balancing air-hole diameter, pitch & gold thickness for max sensitivity



Simulation Time

Large parameter sweeps in COMSOL are computationally expensive



Hardware Integration

Stable WiFi & reliable multi-sensor communication with the ESP32

Machine Learning and the Genetic Algorithm were used precisely to make the optimization step efficient despite these constraints.

CONCLUSION

Two Complementary Achievements

Optical Sensor Design

- Circular PCF-SPR biosensor modified from a Dual-Side Polished design
- FEM (COMSOL) confirms strong SPR excitation & sensing behavior
- AI optimization (ML + Genetic Algorithm) maximizes performance
- Achieves 8000 nm/RIU sensitivity with high resolution

Practical IoT System

- ESP32 + DHT11 + MAX30102 + DS18B20 low-cost prototype
- Real-time data via WiFi to Blynk Cloud & Firebase
- Stable, accurate readings validated through practical testing
- Accessible via mobile app and web dashboard

Future Work



Fabricate & Validate

Build the real PCF-SPR sensor and compare to simulation



New Materials & Geometries

Explore silver, graphene & alternative PCF structures



Deeper AI Models

Apply advanced ML / Deep Learning for faster, more accurate optimization



More Biomedical Sensors

Add ECG, blood pressure & glucose sensing



Dedicated Mobile App

Move beyond Blynk to a custom application



Smart Cloud Alerts

AWS IoT integration with automatic abnormal-reading alerts



Thank You

Questions & Discussion

Smart IoT-Based Healthcare System for Early Disease Detection Using Plasmonic PCF-SPR Sensors

Higher Institute of Engineering and Technology – Kafrelsheikh · July 2026